

Project Report

1. Project Overview

Retail businesses generate large amounts of transactional customer data every day. The objective of this project is to transform raw customer shopping data into meaningful business insights using modern data analytics tools. The project follows an end-to-end analytics workflow involving:

- Data Cleaning
- Feature Engineering
- Database Integration
- Business Analysis
- Dashboard Development

2. Objectives

The project aims to answer important business questions such as:

- Which customer segments contribute the most revenue?
- Which products drive business growth?
- Do discounts increase customer spending?
- Does subscription membership improve revenue?
- Which shipping methods generate the highest sales?
- Which age groups contribute most to total revenue?

3. Technology Stack

Component	Technology
Programming Language	Python
Data Processing	Pandas
Database	MySQL
Query Language	SQL
Visualization	Power BI
Development Environment	Jupyter Notebook, VS Code

4. Data Cleaning

The raw dataset was inspected and transformed to improve quality and consistency.

Steps Performed

Missing Value Treatment

Missing values in the Review Rating column were imputed using category-wise median values to preserve product-specific rating distributions.

```
df.isnull().sum()
✓ 0.0s

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size             0
Color            0
Season           0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64

# Handling null values in ratings column
# Since filling with mean affects the data as there might be chance of outliers, so..
# Filling it with the median value category wise
df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
✓ 0.0s
```

Column Standardization

- Converted column names to lowercase.
- Replaced spaces with underscores.
- Renamed lengthy columns for readability.

Example:

Purchase Amount (USD) became purchase_amount

Duplicate Validation

df.duplicated().sum()

5. Feature Engineering

Two new business-focused features were created.

Age Group

Customers were segmented into quartiles using qcut().

```
# Create a new column age_group using quantiles
labels=['Young Adult','Adult','Middle-aged','Senior-Citizen']
df['age_group']=pd.qcut(df['age'],q=4,labels=labels)
✓ 0.0s
```

```
df[['age','age_group']].head(10)
✓ 0.0s
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior-Citizen
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Business Value

- Enables demographic analysis.
- Supports customer targeting strategies.

Purchase Frequency Days

Text frequencies were transformed into numerical values.

```
# create a column purchase_frequency_days(how often the customer purchases the products)
frequency_mapping={
    'Annually':365,
    'Quarterly':90,
    'Monthly':30,
    'Fortnightly':14,
    'Bi-Weekly':14,
    'Weekly':7,
    'Every 3 Months':90
}
df["purchase_frequency_days"]=df['frequency_of_purchases'].map(frequency_mapping)
Clear All Outputs
```

```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

Business Value

- Supports customer behavior analysis.
- Can later be used for predictive analytics models.

Promotional Feature Optimization

The `promo_code_used` column was removed after verifying that it contained identical information to `discount_applied`.

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

6. Database Integration

```
username = "root"
password = "root"
host = "localhost"
port = "3306"
database = "customer_behaviour"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)

df.to_sql(
    name='customer_shopping',
    con=engine,
    if_exists='replace', # replace existing table
    index=False
)
```

7. Business Analysis

A total of **13 business questions** were answered using SQL.

Major Insights

Revenue by Gender: Male customers generated approximately twice the revenue generated by female customers.

Category Performance: Clothing emerged as the highest revenue-generating category.

Age Group Contribution: Young Adult and Middle-aged customers contributed the largest share of revenue.

Subscription Analysis: Subscribed customers demonstrated higher spending behaviour compared to non-subscribers.

Discount Effectiveness: Discount campaigns increased average purchase amounts.

Shipping Preferences: Express and premium shipping methods generated higher-value transactions.

8. Dashboard Development

The dashboard provides interactive business intelligence capabilities through slicers and visualizations.

KPIs

- Total Customers: **3.9K**
- Average Purchase Amount: **\$59.76**
- Average Review Rating: **3.75**
- Total Revenue: **\$233.08K**
- Subscription Rate: **27%**

Visualizations

- Revenue by Gender
- Revenue by Category
- Revenue by Age Group
- Customer Distribution by Age Group
- Subscription Analysis
- Seasonal Revenue Trends
- Shipping Performance Analysis
- Interactive Filters

9. Business Impact

The analysis enables businesses to:

- Identify high-value customer segments.
- Optimize product inventory.
- Improve discount strategies.
- Increase subscription conversion.
- Optimize shipping operations.